PortfolioTracker.xlsm.VBAProject.RSICalculationModule [Page 1]
Rem Attribute VBA_ModuleType=VBAModule
Option VBASupport 1
Option Explicit
Sub ImportAndCalculateRSI()
 Dim ws As Worksheet
 Dim lastRow As Long
 Dim firstEmptyCol As Long
 Dim period As Long
 Dim firstAvgRow As Long
 Dim changeCol As Long, gainCol As Long, lossCol As Long
 Dim avgGainCol As Long, avgLossCol As Long, rsCol As Long, rsiCol As Long
 Dim rsiRange As Range
 Dim ch As ChartObject
 Dim rsiStartRow As Long
 Dim sheetName As String
 ' Disable screen updating and automatic calculations for performance impro
vement
 Application.ScreenUpdating = False
 Application.Calculation = xlCalculationManual
 ' Step 1: Import stock data
 On Error GoTo ErrorHandler
 Call ImportStockDataFromStooq
 ' Step 2: Use the newly created active sheet
 Set ws = ActiveSheet
 ' Step 3: Read RSI period from the named range "RSITimeFrame"
 On Error Resume Next
 period = CLng(ThisWorkbook.Names("RSITimeFrame").RefersToRange.Value)
 On Error GoTo ErrorHandler
 If period <= 0 Then
 MsgBox "Please set a valid RSI period (positive integer) in the named
cell 'RSITimeFrame'.", vbExclamation
 GoTo Cleanup
 End If
 With ws
 ' Step 4: Remove columns Open, High, Low (B:D), then Volume (new C)
 .Range("B:D").Delete
 .Range("C:C").Delete
 ' Step 5: Find last row of data
 lastRow = .Cells(.Rows.Count, "A").End(xlUp).Row
 ' Step 6: Replace "." with "," in the Close column (column B)
 .Range("B2:B" & lastRow).Replace What:=".", Replacement:=",", _
 LookAt:=xlPart, SearchOrder:=xlByRows
 ' Step 7: Find first empty column on row 1
 firstEmptyCol = .Cells(1, .Columns.Count).End(xlToLeft).Column + 1
 ' Define column positions
 changeCol = firstEmptyCol
 gainCol = changeCol + 1
 lossCol = changeCol + 2

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 avgGainCol = changeCol + 3
 avgLossCol = changeCol + 4
 rsCol = changeCol + 5
 rsiCol = changeCol + 6
 ' Step 8: Add headers
 .Cells(1, changeCol).Value = "Change"
 .Cells(1, gainCol).Value = "Gain"
 .Cells(1, lossCol).Value = "Loss"
 .Cells(1, avgGainCol).Value = "Average Gain"
 .Cells(1, avgLossCol).Value = "Average Loss"
 .Cells(1, rsCol).Value = "RS"
 .Cells(1, rsiCol).Value = "RSI"
 ' Step 9: Fill formulas
 ' Change = Close - Previous Close
 .Range(.Cells(3, changeCol), .Cells(lastRow, changeCol)).FormulaR1C1 =
 _
 "=RC2 - R[-1]C2"
 ' Gain = MAX(Change, 0)
 .Range(.Cells(3, gainCol), .Cells(lastRow, gainCol)).FormulaR1C1 = _
 "=MAX(RC[-1], 0)"
 ' Loss = MAX(-Change, 0)
 .Range(.Cells(3, lossCol), .Cells(lastRow, lossCol)).FormulaR1C1 = _
 "=MAX(-RC[-2], 0)"
 ' Start computing RSI from the row after the full period
 firstAvgRow = 2 + period
 ' Average Gain
 .Range(.Cells(firstAvgRow, avgGainCol), .Cells(lastRow, avgGainCol)).F
ormulaR1C1 = _
 "=AVERAGE(R[-" & (period - 1) & "]C[-2]:RC[-2])"
 ' Average Loss
 .Range(.Cells(firstAvgRow, avgLossCol), .Cells(lastRow, avgLossCol)).F
ormulaR1C1 = _
 "=AVERAGE(R[-" & (period - 1) & "]C[-2]:RC[-2])"
 ' RS = Avg Gain / Avg Loss
 .Range(.Cells(firstAvgRow, rsCol), .Cells(lastRow, rsCol)).FormulaR1C1
 = _
 "=RC[-2]/RC[-1]"
 ' RSI = 100 - 100 / (1 + RS)
 .Range(.Cells(firstAvgRow, rsiCol), .Cells(lastRow, rsiCol)).FormulaR1
C1 = _
 "=100 - 100 / (1 + RC[-1])"
 ' Step 10: Autofit for readability
 .Range(.Cells(1, firstEmptyCol), .Cells(1, rsiCol)).EntireColumn.AutoF
it
 ' Format RSI and RS to 2 decimal places
 .Range(.Cells(2, rsiCol), .Cells(lastRow, rsiCol)).NumberFormat = "0.0
0"
 .Range(.Cells(2, rsCol), .Cells(lastRow, rsCol)).NumberFormat = "0.00"

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 ' Set the start row for RSI (used in the chart later)
 rsiStartRow = firstAvgRow
 End With
 ' === APPLY CONDITIONAL FORMATTING TO RSI COLUMN ===
 With ws
 Set rsiRange = .Range(.Cells(rsiStartRow, rsiCol), .Cells(lastRow, rsi
Col))
 ' Clear existing conditional formats
 rsiRange.FormatConditions.Delete
 ' RSI > 70 = GREEN
 With rsiRange.FormatConditions.Add(Type:=xlCellValue, Operator:=xlGrea
ter, Formula1:="70")
 .Interior.Color = RGB(0, 255, 0)
 End With
 ' RSI < 30 = RED
 With rsiRange.FormatConditions.Add(Type:=xlCellValue, Operator:=xlLess
, Formula1:="30")
 .Interior.Color = RGB(255, 0, 0)
 End With
 ' 30 <= RSI <= 70 = YELLOW
 With rsiRange.FormatConditions.Add(Type:=xlCellValue, Operator:=xlBetw
een, Formula1:="30", Formula2:="70")
 .Interior.Color = RGB(255, 255, 0)
 End With
 End With
 ' Force the recalculation of all formulas (to avoid "Valeur périmée")
 Application.Calculate
 ' Step 11: Add RSI Thresholds legend and interpretation to the right (cell
s, top)
 With ws
 ' Positioning the threshold legend to the right of the data
 .Cells(1, rsiCol + 2).Value = "RSI Thresholds"
 .Cells(2, rsiCol + 2).Value = "RSI > 70 (Overbought)"
 .Cells(2, rsiCol + 2).Interior.Color = RGB(0, 255, 0) ' Green for 70
 .Cells(3, rsiCol + 2).Value = "RSI < 30 (Oversold)"
 .Cells(3, rsiCol + 2).Interior.Color = RGB(255, 0, 0) ' Red for 30
 .Cells(4, rsiCol + 2).Value = "30 <= RSI <= 70 (Neutral)"
 .Cells(4, rsiCol + 2).Interior.Color = RGB(255, 255, 0) ' Yellow for 3
0-70
 End With
 ' Step 12: Create the RSI chart to the right (in rsiCol + 2)
 Set ch = ws.ChartObjects.Add(Left:=ws.Cells(1, rsiCol + 2).Left, Width:=60
0, Top:=100, Height:=300)
 With ch.Chart
 .ChartType = xlLine
 .SetSourceData Source:=ws.Range(ws.Cells(rsiStartRow, rsiCol), ws.Cell
s(lastRow, rsiCol))
 .HasTitle = True
 .ChartTitle.Text = "RSI Over Time"

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 ' Set the X-axis to use dates from column A
 .Axes(xlCategory).CategoryNames = ws.Range(ws.Cells(rsiStartRow, 1), w
s.Cells(lastRow, 1))
 ' Set the minimum and maximum values for the Y-axis
 .Axes(xlValue).MinimumScale = 0
 .Axes(xlValue).MaximumScale = 100
 ' Add a trendline
 .SeriesCollection(1).Trendlines.Add Type:=xlLinear, Name:="RSI Trendli
ne"
 .SeriesCollection(1).Trendlines(1).DisplayEquation = False
 .SeriesCollection(1).Trendlines(1).DisplayRSquared = False
 ' Add the trendline to the legend
 .SeriesCollection(1).Name = "RSI"
 .SeriesCollection(1).Trendlines(1).Name = "RSI Trendline"
 .HasLegend = True
 End With
 ' Construct a compact sheet name with "RSI"
 sheetName = UCase(ws.Name) & "_RSI"
 ' Set the sheet name, handling potential errors
 On Error Resume Next
 ws.Name = Left(sheetName, 31) ' Truncate to 31 characters to fit Excel's s
heet name limit
 On Error GoTo 0
 ' Freeze the first row
 ws.Activate
 ActiveWindow.SplitColumn = 0
 ActiveWindow.SplitRow = 1
 ActiveWindow.FreezePanes = True
 ' Resize columns A to K
 ws.Range("A:K").EntireColumn.AutoFit
 MsgBox "RSI calculation completed on sheet: " & ws.Name, vbInformation
Cleanup:
 ' Re-enable screen updating and automatic calculations
 Application.ScreenUpdating = True
 Application.Calculation = xlCalculationAutomatic
 Exit Sub
ErrorHandler:
 MsgBox "An error occurred: " & Err.Description, vbExclamation
 GoTo Cleanup
End Sub